

Qirui Mi

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🌐 Website: <https://miracle1207.github.io/miqirui.github.io/>

🔍 Google Scholar: <https://scholar.google.com/citations?user=PaRrsDcAAAAJ>

Research Interests

AI: Multi-Agent Systems, Reinforcement Learning (RL) agents, Large Language Model (LLM) agents, Population-agent system, Game Theory.

AI for Economics: Economic simulations, World Model, and Social simulations.

Education

Institute of Automation, Chinese Academy of Sciences, Beijing, China *Aug 2021 – Present*
 PhD Student in Pattern Recognition and Intelligent Systems
 Advisors: Prof. Jun Wang, Prof. Haifeng Zhang

Nanyang Technological University, Singapore *Oct 2024 – Aug 2025*
 Visiting PhD Student in Computer Science
 Advisors: Prof. Bo An

Publications

Think, Speak, Decide: Language-Augmented Multi-Agent Reinforcement Learning for Economic Decision-Making Nov 2025
 Heyang Ma*, **Qirui Mi***, Qipeng Yang, Zijun Fan, Bo Li, Haifeng Zhang
AAAI 2026 (CCF A). [Link](#) [🔗](#)

EconGym: A Scalable AI Testbed with Diverse Economic Tasks Sep 2025
Qirui Mi, Qipeng Yang, Zijun Fan, Wentian Fan, Heyang Ma, Chengdong Ma, Siyu Xia, Bo An, Jun Wang, Haifeng Zhang
NeurIPS 2025 (CCF A). Avg rating: 4.75. [Link](#) [🔗](#)

MF-LLM: Simulating Population Decision Dynamics via a Mean-Field Large Language Model Framework Sep 2025
Qirui Mi, Mengyue Yang, Xiangning Yu, Zhiyu Zhao, Cheng Deng, Bo An, Haifeng Zhang, Xu Chen, Jun Wang
NeurIPS 2025 (CCF A). Avg rating: 4.75. [Link](#) [🔗](#)

Learning Macroeconomic Policies through Dynamic Stackelberg Mean-Field Games Jul 2025
Qirui Mi, Zhiyu Zhao, Chengdong Ma, Siyu Xia, Yan Song, Mengyue Yang, Jun Wang, Haifeng Zhang
ECAI 2025 (CCF B). [Link](#) [🔗](#)

Mean Field Correlated Imitation Learning Apr 2024
 Zhiyu Zhao, Chengdong Ma, **Qirui Mi**, Ning Yang, Xue Yan, Mengyue Yang, Haifeng Zhang, Jun Wang, Yaodong Yang
AAMAS 2025 (CCF B). [Link](#) [🔗](#)

Large Language Models Play StarCraft II: Benchmarks and a Chain of Summarization Approach Sep 2024
 Weiyu Ma, **Qirui Mi**, Yongcheng Zeng, Xue Yan, Runji Lin, Yuqiao Wu, Jun Wang, Haifeng Zhang
NeurIPS 2024 (CCF A). [Link](#) [🔗](#)

TaxAI: A Dynamic Economic Simulator and Benchmark for Multi-Agent Reinforcement Learning

Dec 2023

Qirui Mi, Siyu Xia, Yan Song, Haifeng Zhang, Shenghao Zhu, Jun Wang

AAMAS 2024 (CCF B). [Link](#) 

Joint Caching and Transmission in the Mobile Edge Network: A Multi-Agent Learning Approach

Dec 2021

Qirui Mi, Ning Yang, Haifeng Zhang, Haijun Zhang, Jun Wang

IEEE GLOBECOM 2021 (CCF C). [Link](#) 

Talks

2025 EconGym: A Scalable AI Testbed with Diverse Economic Tasks. Invited talk at *Conference on Machine Learning in Economics*, Peking University, Beijing, China.

2025 Organizing and presenting the Computational Economic Competition. *AAMAS 2025*, Detroit, USA.

2025 Mean Field Correlated Imitation Learning. Oral presentation at *AAMAS 2025*, Detroit, USA.

2025 Game-Theoretic Agents for Economic and Social Governance. Invited talk at *Dadao Forum*, Shandong Institute of Business and Technology.

2024 Game-Theoretic Agents for Economic Problems. Invited talks at *Shanghai Jiao Tong University* (JHC Lecture Series) and *Fudan University*, Shanghai, China.

2024 TaxAI: A Dynamic Economic Simulator and Benchmark for Multi-Agent Reinforcement Learning. Invited talk at *DAI 2024*, Singapore.

2024 TaxAI: A Dynamic Economic Simulator and Benchmark for Multi-Agent Reinforcement Learning. Oral presentation at *AAMAS 2024*, Auckland, New Zealand.

Services

Main organizer of the AAMAS 2025 Computational Economics Competition: Optimal Tax Policy with Heterogeneous Agent Models. [Website](#) 

Main organizer of the AAMAS 2024 Computational Economics Competition: Optimal Tax Policy. [Website](#) 

Main organizer of the CCF Computational Economics Competition. [Website](#) 

Reviewers: TPAMI, NeurIPS, AAAI, AAMAS, DAI.

Honors

Outstanding Student Leader	2024
Outstanding Merit Student	2023
Outstanding Graduate of Beijing Colleges and Universities	2021
Outstanding Undergraduate Thesis of Beijing Colleges and Universities	2021
Meritorious Winner, MCM (USA)	2019
Second Prize, 10th 'Challenge Cup' for Beijing College Students	2019